```
import numpy as np
import pandas as pd
data=pd.read_csv('scaling_data.csv')
print(data)
print("\n")

data.Branch=data.Branch.fillna(data.Branch.mode()[0])
X=data.iloc[:,:-1].values
y=data.iloc[:,-1].values


from sklearn.impute import SimpleImputer
cg=SimpleImputer(strategy="mean",missing_values=np.nan)
stip=SimpleImputer(strategy="mean",missing_values=np.nan)
X[:,[1]]=cg.fit_transform(X[:,[1]])
X[:,[2]]=stip.fit_transform(X[:,[2]])
print(X)
print("\n")

from sklearn.preprocessing import OneHotEncoder
ohe=OneHotEncoder(sparse_output=False)
Branch=ohe.fit_transform(X[:,[0]])
print(Branch)
print('\n')
ready=np.concatenate((Branch,X[:,[1,2]]),axis=1)
print(ready,end = "\n")


from sklearn.preprocessing import StandardScaler
ss=StandardScaler()
std_data=ss.fit_transform(ready)
print(std_data,end = "\n")


from sklearn.preprocessing import MinMaxScaler
mm=MinMaxScaler(feature_range=(0,1))
minmax_data=mm.fit_transform(ready)
print(minmax_data,end = '\n')
```

```
   Branch  CGPA  Stipend Placed
0     CSE   8.9  25000.0    Yes
1     ECE   7.2  12000.0     No
2    MECH   6.8   8000.0     No
3     CSE   9.3  30000.0    Yes
4      IT   8.1  18000.0    Yes
5     ECE   NaN  15000.0     No
6    MECH   7.5  10000.0    Yes
7     CSE   8.4      NaN    Yes
8      IT   6.9   9000.0     No
9     NaN   7.8  14000.0     No
```

```
10    ECE   8.6  20000.0    Yes
11     IT   7.1  11000.0     No
```

```
[['CSE' 8.9 25000.0]
 ['ECE' 7.2 12000.0]
 ['MECH' 6.8 8000.0]
 ['CSE' 9.3 30000.0]
 ['IT' 8.1 18000.0]
 ['ECE' 7.872727272727272 15000.0]
 ['MECH' 7.5 10000.0]
 ['CSE' 8.4 15636.363636363636]
 ['IT' 6.9 9000.0]
 ['CSE' 7.8 14000.0]
 ['ECE' 8.6 20000.0]
 ['IT' 7.1 11000.0]]
```

```
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 [0. 0. 0. 1.]
 [1. 0. 0. 0.]
 [0. 0. 1. 0.]
 [0. 1. 0. 0.]
 [0. 0. 0. 1.]
 [1. 0. 0. 0.]
 [0. 0. 1. 0.]
 [1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]]
```

```
[[1.0 0.0 0.0 0.0 8.9 25000.0]
 [0.0 1.0 0.0 0.0 7.2 12000.0]
 [0.0 0.0 0.0 1.0 6.8 8000.0]
 [1.0 0.0 0.0 0.0 9.3 30000.0]
 [0.0 0.0 1.0 0.0 8.1 18000.0]
 [0.0 1.0 0.0 0.0 7.872727272727272 15000.0]
 [0.0 0.0 0.0 1.0 7.5 10000.0]
 [1.0 0.0 0.0 0.0 8.4 15636.363636363636]
 [0.0 0.0 1.0 0.0 6.9 9000.0]
 [1.0 0.0 0.0 0.0 7.8 14000.0]
 [0.0 1.0 0.0 0.0 8.6 20000.0]
 [0.0 0.0 1.0 0.0 7.1 11000.0]]
[[ 1.41421356e+00 -5.77350269e-01 -5.77350269e-01 -4.47213595e-01
   1.32236874e+00  1.46452077e+00]
 [-7.07106781e-01  1.73205081e+00 -5.77350269e-01 -4.47213595e-01
  -8.65975986e-01 -5.68745929e-01]
 [-7.07106781e-01 -5.77350269e-01 -5.77350269e-01  2.23606798e+00
  -1.38088063e+00 -1.19436645e+00]
```

```
 [ 1.41421356e+00 -5.77350269e-01 -5.77350269e-01 -4.47213595e-01
   1.83727338e+00  2.24654642e+00]
 [-7.07106781e-01 -5.77350269e-01  1.73205081e+00 -4.47213595e-01
   2.92559455e-01  3.69684854e-01]
 [-7.07106781e-01  1.73205081e+00 -5.77350269e-01 -4.47213595e-01
   0.00000000e+00 -9.95305376e-02]
 [-7.07106781e-01 -5.77350269e-01 -5.77350269e-01  2.23606798e+00
  -4.79797506e-01 -8.81556190e-01]
 [ 1.41421356e+00 -5.77350269e-01 -5.77350269e-01 -4.47213595e-01
   6.78737935e-01 -2.84499275e-16]
 [-7.07106781e-01 -5.77350269e-01  1.73205081e+00 -4.47213595e-01
  -1.25215447e+00 -1.03796132e+00]
 [ 1.41421356e+00 -5.77350269e-01 -5.77350269e-01 -4.47213595e-01
  -9.36190256e-02 -2.55935668e-01]
 [-7.07106781e-01  1.73205081e+00 -5.77350269e-01 -4.47213595e-01
   9.36190256e-01  6.82495115e-01]
 [-7.07106781e-01 -5.77350269e-01  1.73205081e+00 -4.47213595e-01
  -9.94702147e-01 -7.25151060e-01]]
[[1.         0.         0.         0.         0.84       0.77272727]
 [0.         1.         0.         0.         0.16       0.18181818]
 [0.         0.         0.         1.         0.         0.        ]
 [1.         0.         0.         0.         1.         1.        ]
 [0.         0.         1.         0.         0.52       0.45454545]
 [0.         1.         0.         0.         0.42909091 0.31818182]
 [0.         0.         0.         1.         0.28       0.09090909]
 [1.         0.         0.         0.         0.64       0.34710744]
 [0.         0.         1.         0.         0.04       0.04545455]
 [1.         0.         0.         0.         0.4        0.27272727]
 [0.         1.         0.         0.         0.72       0.54545455]
 [0.         0.         1.         0.         0.12       0.13636364]]
```